

spaghet boi's jong note (part of)

June 10, 2024

Contents

1	Tile Efficiency	1
1.1	Complex Taatsu	2
1.1.1	Ryankanchan and Its Derived Shapes	7
1.1.2	134 Shape	16
1.1.3	When the 5 of 578 Shape is an Akadora	17
1.1.4	3568 Shape	18
1.1.5	35689 Shape	20
1.1.6	Sankanchan and Yonkanchan	20
1.1.7	Shapes with Overlapping Acceptance	21
1.2	Shapes Involving Mentsus	22
1.2.1	Sanmenchan	22
1.2.2	Yonrenkei	23
1.2.3	23445 Shape	25
1.2.4	Ryanmenkanchan	26
1.2.5	1345 Shape	28
1.2.6	13457 Shape	28
1.2.7	Entotsu	28
1.3	6 Block Theory	29
1.3.1	Standard of Keeping 6 Blocks	29
1.3.2	Order of Dropping a Block	31
2	Bibliography	33

1 Tile Efficiency

Tile efficiency is a set of policies and techniques that allows us to reach an optimal point when taking speed, value, win rate, and many other factors into consideration. There is also the concept of *pure* tile efficiency which concerns of only the speed of reaching *tenpai*, one tile away from completing our hand. As we will see later, focusing solely on pure tile efficiency will lead us to adverse situations.

The following text assumes that the reader has a basic understanding of mahjong rules such as

suits, tiles, win conditions, call mechanisms, and riichi. There are, however, many terms that I cannot find a proper translation in English, so I will use Japanese terms as is. They are listed in Table 1 along with some mahjong jargons. Many other terms such as *ryanmenkanchan* and *atosuji* will be introduced later in the text.

Term	Definition
Ryanmen	The shape consisting of two neighboring number tiles such as  (23p)
Kanchan	The shape consisting of tiles whose numbers have difference of 2 such as  (24p)
Penchan	The shape consisting of number tiles of 1 or 9 and a neighboring tile such as  (12p)
Toitsu	The shape consisting of two identical copies such as  (11p)
Taatsu	The set of ryanmen, kanchan, and penchan
Shuntsu	The shape consisting of three consecutive number tiles such as  (123p)
Koutsu	The shape consisting of three identical copies such as  (111p)
Minkou	A koutsu resulting from pon call and thus open to all players
Ankou	A koutsu in our hand invisible to other players
Mentsu	The set of koutsu and shuntsu, completed shapes
Tenpai	One tile away from completing our hand
1-away	One tile away from tenpai
2-away	Two tiles away from tenpai
Shanpon	A wait pattern where we have two pairs in our hand

Table 1: Terminology

The following text is an excerpt of my mahjong notes translated into English. This is the part of my notes that, based on my experience, that makes huge difference for players that have basic understanding of tile efficiency but get confused when encountered with complex shapes. I organized this part into three subparts: complex taatsu, taatsus involving mentsus, and 6 block theory. This is by no means all of tile efficiency; there are more complex shapes, more difficult table situations, more specialized topics out there. Mahjong is a long journey.

There is also this notation $\frac{3}{12}$ which is my shorthand for “3 kinds 12 tiles.” It will be used in tables comparing the acceptance or improvement of drops and it will be followed by such tiles.

1.1 Complex Taatsu

Complex taatsus are shapes involving simple taatsus (penchans, kanchans, and ryanmens) and an additional tile. Table 2 lists complex taatsus that uses three tiles.

Name	Shape	Acceptance
Penchan-toitsu		 6 tiles
Kanchan-toitsu		 6 tiles
Ryankanchan		 8 tiles
Ryanmen-toitsu		 10 tiles

Table 2: Types of complex taatsus using three tiles

Since complex taatsus except for ryankanchan contain a pair, they must serve as head when there is no other head candidate in our hand. On the other hand, when we have more than two complex taatsus containing pairs, the pairs will impair our acceptance, so in most cases we opt to break those pairs and fix them as single taatsus. In order to increase the probability of reaching good waits, as a rule of thumb, we drop one tile from the strongest complex taatsu (ネマタ, 2014g,h). Table 3 lists our actions based on the number of pairs, complex taatsus (excluding ryankanchans), and ryanmen-toitsus in our hand.

Pair	Complex taatsu	Ryanmen-toitsu	Action
2	2		Fix the weakest complex taatsu into a pair
3	2	0	Break the pair in the inner kanchan-toitsu Fix outer penchan-toitsu into a pair
3	2	1	Fix the ryanmen-toitsu into a ryanmen
3	3	0	Fix the weakest complex taatsu into a pair
3	3	1	Fix the ryanmen-toitsu into a ryanmen
3	3	2	Fix the weakest complex taatsu into pair

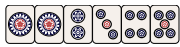
Table 3: Action based on the number of pairs, complex taatsus, and ryanmen-toitsus.

Two Pairs

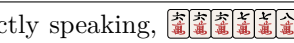
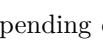
We must drop a tile from either  or  from (1) since we will not be touching completed groups and simple ryanmens. To keep two pairs within our hand, our choices narrow down to  and . Their acceptances are listed in Table 4, which suggests that dropping  has 4 more tiles than dropping  in terms of acceptance. This is a case of fixing the weaker complex taatsu into a pair and keeping the stronger complex taatsu.

(1)	
	Drop  
	Acceptance $6/20$:  $5/16$: 

Table 4: Acceptance of (1).

(2) has  which makes dropping  and fixing  ryanmen attractive. But by doing so, we lost flexibility in manzu shape and we must make either  or  our head, thus reducing our acceptance. If we see (2) as choosing between ¹ and , it becomes clear that dropping  here gives us the wide acceptance. Now, dropping  also gives the same acceptance as , but considering the case when we draw  and wait on , dropping  here makes the outside tile  more likely to come out. Hence drop  (渋川難波, 2024). Table 5 lists our acceptance.

(2)	East-1, East seat, turn 6, dora 
	

¹Strictly speaking,  can be treated as either  +  or  + , depending on which mentsu we extract from it. Either way, manzu shape is made of a ryanmen-toitsu and a mentsu; it does not change the fact that we are comparing between a ryanmen-toitsu and a penchan-toitsu.

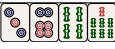
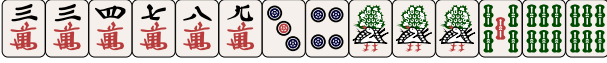
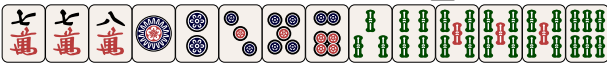
Drop		
Acceptance	4/15: 	8/24: 

Table 5: Acceptance of (2).

If our choices are between two ryanmen-toitsus, fix the inner one into a pair to make sure that when we reach tenpai our waits are outer, which has higher win rate. Therefore, between  and  from (3), we cut  (お知らせ, 2020, pp. 208–209).

(3) 

(4) also has two ryanmen-toitsus   and  , and one is connected to a completed mentsu . Upon seeing this complex shape, one may be reluctant to touch complex souzu shape and tempted to drop  to fix   ryanmen or drop . The  drop forces us to treat   as our head and renders  surplus, thus having less acceptance than  drop, while  drop leads us to final wait on either  or . Our previous conclusion applies here as well: drop  since we have  which is inside (渋川難波, 2024). Table 6 lists our acceptance.

East-1, East seat, turn 6, dora 
 (4) 

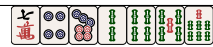
Drop			
Acceptance	4/16: 	7/22: 	7/23: 

Table 6: Acceptance of (4).

It should be noted that in all cases with only two pairs, fixing complex taatsus into simple taatsus will not lead to maximal acceptance. When we have two pairs in our hand, we can accept tiles of the same kind as these pairs; any tiles that follows these pairs also increases acceptance. If we only have one pair, say  , then the tile that follows the pair becomes useless, since drawing  destroys the pair and produces a mentsu, which renders our hand headless and unready. This means that sometimes we have to counterintuitively destroy ryanmens.

Three Pairs, Two Complex Taatsus

Since we have more than two pairs in (5), we opt to break one, either from  or ². Breaking the pair in both complex taatsus will give us the same number of acceptance, but the numbers of draws that lead to ryanmen tenpai are different and listed in Table 7. This is a case of fixing stronger complex taatsu into a simple taatsu (ネマタ, 2014g).

(5) 

If our choices are between complex taatsus without ryanmens, there is no difference in direct acceptance and ryanmen-tenpai draws. If our complex taatsus contains kanchans, there is a chance

²Note that we do not drop  here since that is equal to dropping entire block and increasing our shanten number.

Drop		
Acceptance	5/16:	5/16:
Ryanmen-tenpai draws	3/8:	1/4:

Table 7: Acceptance of (5).

for those to upgrade into ryanmens. It is therefore optimal to break the pair from inner kanchan-toitsus, since they have 8 tiles of ryanmen improvement as opposed to 4 tiles for outside kanchans. This means dropping from in (6). Drawing improves our kanchan into ryanmen, but only can produce a ryanmen taatsu out of (ネマタ, 2014g).

(6)

For complex taatsus with few ryanmen-improvement draws, if the remaining pair is on the inner part, we can expect its ryanmen improvement. We need to fix a complex taatsu into a pair for that outcome, and it then becomes optimal to fix the weakest complex taatsu into a pair. (7) has that, when turned into a penchan (simple taatsu), has no ryanmen improvement, compared to which produces a ryanmen when we draw , so we drop and make it a pair (ネマタ, 2014g).

(7)

(8) has a more complex shape that can be treated as either + or + + , which leads to its wide acceptance of . We will lose some of its acceptance if we drop any tile from this shape, so instead we fix into a ryanmen, since technically we still have three pairs in our hand. Table 8 lists the acceptance after different drops. This leads to the rule of thumb that we keep the more complex shape with more acceptance intact (G・ウザク, 2016, p. 24).

(8) East-1, North seat, turn 7, dora

Drop			
Acceptance	8/24:	7/21:	6/19:

Table 8: Acceptance of (8).

(9) is also choosing between two ryanmen-taatsus and , and fixing either into a ryanmen does not make any difference in terms of acceptance. However, if we keep and draw another dora , we can drop and go for tanyao. In that regard, it is best that we drop here (G・ウザク, 2016, p. 24).

(9) East-1, East seat, turn 6, dora 

If there are three complex taatsus, all of which contain no ryanmen, then it is optimal to fix the one with least ryanmen improvement into a pair, so that when the remaining complex taatsus becomes a mentsu, there will be no surplus tiles. (11) is the hand derived from (10) by dropping and completing . (12) is derived from (10) by dropping instead and completing . We can see, in Table 9, that dropping loses acceptance and we have 2 tiles of acceptance less than dropping , but once becomes a mentsu, (11) has 4 tiles of acceptance more than (12). We therefore prioritize the width of 1-away acceptance and sacrifice a little bit at 2-away. Drop (ネマタ, 2014g).

(12) 

Drop			
Acceptance	7/22:       	7/24:       	
Hand after completing		(11)	(12)
Acceptance after completing		5/16:     	3/12:   

Table 9: Acceptance of (10), (11), and (12)

If there is one ryanmen-toitsu among all three complex taatsus, it is optimal to fix the ryanmen-toitsu into a ryanmen. There will be no difference in immediate acceptance between the drop candidates from (13), but there will be huge differences when we draw any of the acceptances, as shown in Table 10. If we drop , completing the remaining two complex taatsus and will leave us only 3 kinds 12 tiles of acceptance. Likewise for drop. On the other hand, if we drop , all effective draws will guarantee us 4 kinds 20 tiles of acceptance. One may be tempted by the draw after dropping which gives us complete iishanten, but also keep in mind that it is more likely to draw ourselves into 3 kinds 12 tiles of acceptance.

Since it is difficult to calculate all the acceptance of all possible effective draws after dropping each possible tiles, we can just follow the principle to reduce the strongest complex taatsu to a ryanmen.

Drop	Next draw	Max acceptance
		drop → 5/16:
		drop → 5/16:
		drop → 4/16:
		drop → 4/16:
		drop → 4/16:
		drop → 4/16:
		drop → 5/16:
		drop → 3/12:
		drop → 3/12:
		drop → 6/20:
		drop → 3/12:
		drop → 3/12:

Table 10: Differences between cutting and from (13)

Three Pairs, Three Complex Taatsus, Two Ryanmen-Toitsus

If there are two ryanmen-toitsus, then we fix the remaining complex taatsu into a pair. After that, since we have three pairs in our hand, we will have 2 tiles of acceptance less than fixing the other taatsus into a ryanmen. Considering the acceptance when we reach 1-away, however, it becomes clear that this sacrifice of 2 tiles at 2-away is worth. If we choose to fix any ryanmen in (14), completing any complex taatsus leaves us 1 surplus tile. On the contrary, by fixing the weakest complex taatsu as our head, we are guaranteed complete iishanten no matter what we draw. Drop , even though it may be a little bit counterintuitive because we have been hearing that three pairs are the weakest.

(14)

1.1.1 Ryankanchan and Its Derived Shapes

The shape is called *ryankanchan* in Japanese, sometimes *ryankan* for short. The shape contains two kanchans and , and *ryan* in Japanese means “two”, hence the name. It takes up the space of 3 tiles and accepts at most 8 tiles, and we are left to chose any of its two kanchans when other parts of our hands get filled up, so ryankanchan is weaker than ryanmen in terms of space efficiency and final waits.

Comparison between Ryankanchans

Even between different kinds of ryankanchans, there are differences other than the tiles used. Table 12 lists three kinds of ryankanchan , , and . Note that we omit and because they are identical to and due to symmetry of mahjong tiles. We can see that has the most improvement draws, and dropping from makes our suji wait better than non-suji kanchan wait. and the ryanmen developed from it are all tanyao tiles, so it is a shape we would like to keep for tanyao. However, dropping a tile from

Drop	Next draw	Max acceptance
	 	drop $\rightarrow 6/20$: drop $\rightarrow 4/16$: drop $\rightarrow 4/16$: drop $\rightarrow 4/16$:
	 	drop $\rightarrow 6/20$: drop $\rightarrow 6/20$: drop $\rightarrow 6/20$: drop $\rightarrow 6/20$: drop $\rightarrow 6/20$: drop $\rightarrow 6/20$: drop $\rightarrow 6/20$:

two ryankanchans will largely depend on the actual table situation.

Table 12: Comparison between ryankanchans

(15) has a ryankanchan and an ankou. If we drop  and make  our head, we will have only 4 kinds 16 tiles of acceptance at 1-away, as opposed to 7 kinds 24 tiles if we drop  while still maintaining 1-away, as shown in Table 13. Besides the number of acceptance, when we keep  , any draw of  develops our souzu shape, and any draw of manzu that gives us head leaves us ryanmen tenpai. The problem becomes which of  to drop. Considering the acceptance of  and the improvement draw , drop  (G・ウザク, 2019, p. 84).

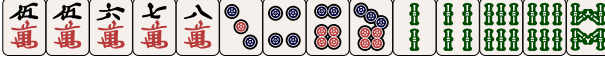
Drop   

Acceptance $7/24$:        $7/24$:          $4/16$:    

Since (16) has two ryanmens with overlapping acceptance, dropping ryanmens and dropping any of  from  will not lead to any difference in terms of the number of acceptance. However, considering the shape of our final wait, we exclude ryanmens from our drop candidates. We must keep  pair in case we draw any of  which produces ryanmen. We also cannot

overlook the possibility of 678 sanshoku, so we are incentivized to  keep. In this regard,  becomes the most unwanted tile. Drop  (G・ウザク, 2016, p. 14).

East-1, East seat, turn 5, dora 

(16) 

Drop			
Acceptance	7/24: 	6/24: 	7/24: 

Table 14: Acceptance of (16).

When the 5 of 135 Ryankanchan is Akadora

The draws that are related to the akadora of  ryankanchan are , which makes  seems useless. That said, when we draw ,  becomes floating and  draws makes  useful. Besides, when we draw another , it produces a head and a kanchan; drawing  further develops it into a ryanmen. Therefore, we would like to keep it intact, unless we have enough blocks to aim for tenpai (ゆうせー, 2024b).

For example, in (17) since we only have  +  +  +  four blocks, we must drop  and aim for draws that give us another block. If we draw , as in (18), then we can drop  and expect blocks around floating  and .

East-1, turn 1, dora 

(17) 

East-1, turn 2, dora 

(18) 

(19) is a similar hand, but since we already have 5 blocks, we can drop  and go for tanyao.

East-1, turn 2, dora 

(19) 

The  shape in (20) is the predecessor of  ryankanchan, so it will be a loss if we drop  over otakaze . However, this possibility will not make  stronger than seat wind or yakuhai . So we drop  for now, and drop  later on.

East-2, North seat, dora 

(20) 

Vertically Developed Ryankanchans such as 2246 and 2466

When we have and draw another copy of , the result shape is a strong shape that accepts 4 kinds 12 tiles. Likewise for and drawing , and and drawing . If we draw both , we end up the shape which is called *ryanmenkanchan*³ (chirno, 2008). Despite all these improvements, we still have to consider the cases where we have more stronger taatsus and must drop one from or .

We can drop either or from (21), which will lead to the same number of 16 tiles of acceptance, as shown in Table 15. Keeping two pairs in our head does not give us more ryanmen improvement draws than keeping shape (both having only , and drawing any of these pairs into an ankou leaves rii-nomi. In that regard, it is slightly more advantageous to keep shape and hope for pinfu. Drop (G・ウザク, 2019, p. 86).

(21)

	Drop		
Acceptance	4/16:		5/16:
Pinfu draws	2/8:		1/4:

Table 15: Acceptance of (21).

(22) has instead of compared to (21), which increases the draws that give us complete iishanten from to . In that regard, it is more advantageous to keep two pairs rather than keeping ryankanchan. Drop (G・ウザク, 2019, p. 88).

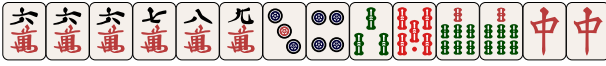
(22)

Dropping and from (23) leaves us the same number of acceptance. However, if we drop and have drawing means that is very to become unusable and we are likely to end up rii-nomi waiting on kanchan wait. Even the ryanmen improvement means that we have to give up . In this regard, keeping shape has more value and makes our kanchan wait more worth. Drop (G・ウザク, 2016, p. 26).

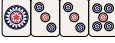
(23) East-1, East seat, turn 7, dor

Since we have a in (24), when we call pon on we must have a pair elsewhere. In that regard, we must keep pair and drop . It is likely that becomes useless, but do keep in mind that open hands are meant to be fast while sacrificing hand value.

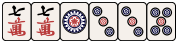
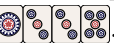
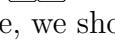
³See Section 1.2.4 for more details on ryanmenkanchan.

(24) East-1, West seat, turn 6, dora 


Bulging Ryankanchans such as 1335 and 2446

To handle shapes like  and .

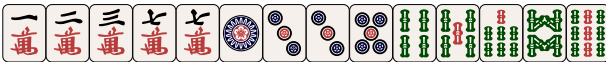
- 2-away
 - Drop outer tile, aim for ryanmen improvement
- 1-away
 - If pinfu is within sight, cut the bulging part
 - If pinfu is infeasible, cut the inner tile

Suppose that a part of our hand is  and the rest of our hand being completed blocks and a ryanmen. Table 16 lists the pinfu draws, ryanmen improvement draws, and final waits (when the ryanmen is completed) of different drops from . When we reduce  to a regular ryankanchan, it has the most draws that give us pinfu; if we cannot draw  in time, we can still cut  and riichi and make our  wait a suji. When pinfu is not feasible, we should cut  in advance to make our suji wait  more likely to come out⁴. When our hand is callable, the  pair becomes more valuable.

Drop			
Result shape			
Pinfu draws	1/4: 	2/8: 	1/4: 
Ryanmen improvement draws	3/12: 	None	2/8: 
Final waits	 shanpon	 kanchan	 kanchan  shanpon

Table 16: Comparing drops from the shape .

(25) gains pinfu if we drop  and draw . Meanwhile, (26) cannot have pinfu no matter what we drop; in this case, drop  and make  more likely to come out. (27) has no pinfu either, and it has  ankou, which means that, after dropping , calling pon on both  will give us tenpai (ゆうせい, 2021c).

(25) East-1, East seat, turn 5, dora 


(26) East-1, East seat, turn 5, dora 


⁴The argument that the suji of a tile cut in advance is more likely to come out than the suji of a riichi declaration tile is based on the deal-in rate difference of sujis. Sujis of riichi declaration tiles have higher deal-in rates than other kinds of sujis (みーにん, 2017, pp. 152–153). If a player's policy is to cut the suji with lower deal-in rate, they will tend not to cut riichi declaration tile suji.

Draw	Result shape	Composition
		Shuntsu + ryankanchan
		Ankou + kanchan
		or ryanmen-toitsu + ryankanchan
		Ryanmenkanchan (see 1.2.4)
		Ryanmen-toitsu + shuntsu

Table 17: Draws that produce a mentsu out of shape.

Since dropping any tile from annihilates the benefits mentioned earlier, when compared to yonrenkei or nakabukure, we tend to keep and fix other 4-tile shapes into a mentsu. As shown in Table 18, dropping either of also gives us far more acceptance than dropping any tile from . Therefore, drop from (31).

(31)	
	Drop
	Acceptance $\frac{3}{12}$: $\frac{2}{8}$: $\frac{5}{16}$:

Table 18: Acceptance of (31).

When we have 6 blocks, we still tend to keep shape. When there are only ryanmens in the rest of our hand, drop . In this regard, since there is a weaker kanchan in (32), we drop ; since there is no weaker taatsu in (33), drop .

(32)	
(33)	

When we have 5 blocks, we still tend to keep shape. When there is no head in our hand, how we handle shape will largely depend on table situation and turn number. When we are still early in the game, we can drop and focus on ryanmen. When we are late and there is no room for improvement, make our head and drop . Therefore, even for the same hand such as (34), our decision changes depending on other information.

(34)	
------	--

Dropping from in advance at 1-away is not recommended (ゆうせー, 2021a). When we drop from (35), our ryanmen-tenpai draw reduces to only , opposed to if we keep it, while both options have the same number of bad-wait draws, as shown in Table 19. Keeping a safe tile here worsens our wait quality too much.

(35) 

Drop		
Result shape		
Ryanmen-tenpai draws	1/4: 	2/8: 
Bad-wait draws	2/8: 	3/8: 

Table 19: Acceptance of (35).

That said, we can drop the ryankanchan part when we draw . Even though the shape  has overlapping acceptance , our bad-wait draws reduce to  3 kinds 6 tiles, 2 tiles less than keeping , as shown in Table 20. We also gain space for safe tiles. Therefore, drop  and then  from (36).

(36)

Drop	
Result shape	
Ryanmen-tenpai draw	2/8: 
Bad-wait draw	3/6: 

Table 20: Acceptance of (36).

The same principle applies to inner  shape (ゆゝせ一, 2022c). Drop  from (37) and (38).

[illegible]

(38) 此此此

Another case where we destroy $\begin{smallmatrix} \text{一} & \text{三} & \text{四} & \text{四} & \text{六} \\ \text{一} & \text{三} & \text{四} & \text{四} & \text{六} \end{smallmatrix}$ shape at 1-away is when we have a 3-7 pair elsewhere and we draw $\begin{smallmatrix} \text{一} \\ \text{一} \end{smallmatrix}$. When we draw $\begin{smallmatrix} \text{一} \\ \text{一} \end{smallmatrix}$ and drop $\begin{smallmatrix} \text{四} \\ \text{四} \end{smallmatrix}$, we end up a kanchan-toitsu, a ryanmen, and a pair. This gives us the same number of acceptance, but the number of ryanmen-improvement draws increases, as shown in Table 21. Therefore, drop $\begin{smallmatrix} \text{四} \\ \text{四} \end{smallmatrix}$ from (39).

(39)

Drop		
Result shape		
Ryanmen-tenpai draws	$\frac{2}{8}$:	$\frac{2}{8}$:
Bad-wait draws	$\frac{2}{8}$:	$\frac{2}{8}$:
Ryanmen-improvement draws	$\frac{2}{8}$:	$\frac{1}{3}$:

Table 21: Acceptance of (39).

When we have yakuhai ankou, we can still drop and keep to make our pon calls more likely. Drop from (40).

(40) East-1, West seat, turn 6, dora

Detached Ryankanchan such as 245679

The shape in (41) is dubbed “detached ryankanchan” since it accepts far apart. If we draw , we can further improve it into a ryanmenkanchan. On the other hand, dropping either will leave us a mentsu and kanchan; which to drop largely depends on the rest of our hand.

In (41), dropping either of from detached ryankanchan or dropping from will leave us the same number of acceptance at 1-away. However, considering the draws that will give us pinfu and the sanshoku chance, fixing ryanmen here is more advantageous, as shown in Table (41) (G・ウザク, 2019, p. 89).

(41)

Drop			
Acceptance	$\frac{4}{16}$:	$\frac{5}{16}$:	$\frac{5}{16}$:
Pinfu draws			
Sanshoku-chance draws			None

Table 22: Acceptance of (41).

Two ryankanchans

(42) has two ryankanchans, one bulging and one vertically developed on one end. Since there is no pair in the rest of our hand, we will focus on how we can make tiles following and pair function. Dropping makes useless since completing shape leaves us no pair and taatsu combination. Dropping either of will make full use of all tiles while giving us the same number of acceptance. Taking ryanmen improvement into consideration, drop becomes optimal, as shown in Table 23.

(42)

Drop			
Acceptance	4/12:	3/12:	2/8:
Ryanmen-improvement draws	2/8:	3/12:	2/8:

Table 23: Acceptance and improvement of (42).

When we must choose between two simple ryankanchans, drop the tile that will make our wait better (Feng, 2018, p. 26) Thus, we drop from (43) and make our wait suji.

(43)

1.1.2 134 Shape

It is a common pitfall to only see the ryanmen part of a shape and drop early. However, the possibilities of can help us tremendously especially when we are early and our hand lacks shapes (ヨーター, 2024). In short,

- Makes our hand *kuttsuki*⁷
- Easy to produce a pair

Draw	Shape	Comment
		A pair and a ryanmen.
		A yonrenkei, which produces a pair easily or functions as nobetan when we reach tenpai.
		A kanchan and a ryanmen.
		A kanchan and a pair. Decent as our final wait.
		A floating and a mentsu. However, drawing gives us ryanmen, drawing also gives us a decent kanchan. Drawing and we have the shape which can easily further develop into a mentsu and a ryanmen.
		Two kanchans.

Table 24: Development of .

Therefore, when our hand does not have 5 blocks ready, we keep over single yakuhais. When there is no head in our hand, we also prioritize . Otherwise, dropping is fine.

(44) does not have enough blocks, so we keep and expect it to develop. Drop from (44).

(44) East-1, West seat

⁷*Kuttsuki* is a Japanese term whose literal meaning is “sticky,” In mahjong context, it refers to a state where tiles next to or 1 tile away from floating tiles becomes acceptable.

(45) already has 5 blocks to aim for tenpai, so we can drop and keep as safe tiles.

East-1, West seat

(45)

(46) has no head confirmed, so we keep to increase our chances of getting a head. Drop .

East-1, West seat, dora

(46)

Play Defensively when Our Hand is Too Messy

(47) is 5-away and only has one ryanmen . No matter how good can develop, it is more likely that some one will slam riichi earlier than us do. In this case, we drop and keep safe tiles .

(47)

Avoid No Dora Bad-wait Rii-Nomi

(48) has the same tiles as (46) but dora is instead of . The possible kanchan that brings us becomes unfavorable, so we drop and fix it into a ryanmen.

East-1, West seat, dora

(48)

When the 1 is Also a Dora

As shown in Table 25, when we have no akadora in (49), keeping dora or taking ryanmen does not make too much difference in terms of hand value. Hence drop and take ryanmen to increase win rate. When we have one or more akadoras, keeping dora , and possibly calling pon and pivot to tanki or drawing another and waiting on shanpon greatly increase hand value. When it is still early, with akadoras in our hand, it is advantageous to drop (ゆうせい, 2022i).

East-1, West seat, dora

(49)

1.1.3 When the 5 of 578 Shape is an Akadora

When we are still far away from tenpai, we can expect to draw tiles around . Since (50) has the risk of ending up no dora bad-wait rii-nomi, we drop from . (51) also has the same risk, and has more ryanmen-improvement draws than , so we drop (ゆうせい, 2022e).

	Drop	Drop
+	1000 points: 7 tiles; 2000 points: 1 tile	2000 points
+	2000 points: 7 tiles; 3900 points: 1 tile	3900 points
+	3900 points: 7 tiles; 7700 points: 1 tile	7700 points

Table 25: Value of (49) depending on drop and draw.

Hand state	Action
Far away from tenpai	Keep
1-away	Drop if tanyao is infeasible
Tenpai	Generally drop

Table 26: Handing .

(50)

(51)

When we are 1-away, acceptance and final wait becomes more important than the value the dora brings. (52) has yaku which gives us value of 1 han and we do not have to cling to for value. Drop

(52)

However, as an exception, when keeping gives us sanshoku or tanyao which enables us to call and speed up our hand, we drop . When reaching tenpai, if dropping guarantees sanshoku and tanyao, we can also dama our hand instead of dropping and riichi. In this regard, (53) is at tenpai but does not have sanshoku confirmed, we drop and riichi; on the other hand, (54) has confirmed sanshoku, and its value is already mangan, we drop but do not riichi.

(53)

(54)

1.1.4 3568 Shape

It is easy to produce a head out of shape (ゆうせー, 2022b). Not just stacking , but drawing also gives us yonrenkei. Even as a floater, is also strong. Thus, it is generally treated as 2 blocks. Table 27 lists our policy when taking the rest of our hand into consideration.

(55) already has ankou, and directly accepting using penchan is valuable. Thus, we drop first, and then drop if it does not give us ryanmen.

Number of block	Compared with	Action
4		Keep
5	Floater (no head in hand)	Keep
	Floater tiles other than 34567	Keep
	Penchan (bottleneck of tanyao)	Keep
	Penchan (not bottleneck of tanyao)	Drop
	Floater tiles of 34567	Drop
6		Drop

Table 27: Handling or (ゆうせー, 2022b; クリアレイン, 2023)

(55) East-1, West seat, turn 3, dora

1.1.5 35689 Shape

Generally, drop 3 at 1-away. With

- Drawing  produces a ryanmen with overlapping acceptance
- Drawing  does not give us a ryanmen; it produces a mentsu  and a kanchan  and a surplus tile 

The only two cases where we drop penchan  and go back to 2-away are (ゆゝせー, 2022h):

1. Penchan is the bottleneck of tanyao
2. We have other strong 3–7 floaters

When we are comparing between floaters in (59) at 2-away,  from  is inferior to isolated . Therefore, drop .

[illegible]

Likewise, for (60) which is 1-away, we keep  penchan and cut  floater.

(60) East-1, West seat, turn 5, dora 



If we later draw  and get *yonrenkei*  in (61), we can drop  penchan. Since this penchan still has overlapping acceptance, when we have very strong shapes in the rest of our hand, we can opt to drop penchan.

(61) East-1, West seat, turn 6, dora 



If this  penchan is the only bottleneck of tanyao, we can drop it in exchange of value and flexibility even if it increases our shanten number. Drop  from (62).

[illegible]

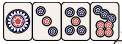
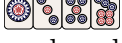
1.1.6 Sankanchan and Yonkanchan

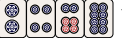
Shapes like  are dubbed *sankanchan* and  *yonkanchan* in Japanese, because they have three kanchans and four kanchans respectively. In general, the order of importance is “even-

Drop		
Acceptance	$3/12$: 	$3/12$: 
Ryanmen-improvement draws	 8 	

Table 28: Acceptance of .

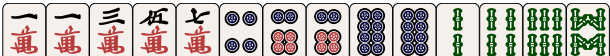
numbered sankanchan < floaters < odd-numbered sankanchan < outer kanchan < yonkanchan < inner kanchan” (chirno, 2008).

For odd-numbered sankanchan like , drawing  produces a ryankanchan and a head, drawing  produces a yonkanchan, drawing  produces a ryankanchan and a kanchan, drawing  produces a ryanmen, and drawing  completes a mentsu. Therefore, the acceptance of  is . However, if we must choose between an outside kanchan and a sankanchan, dropping  from  while keeping the kanchan gives us the same number of acceptance but more ryanmen-improvement draws, as shown in Table 28. Hence the conclusion that sankanchans are weaker than outside kanchans.

Yonkanchans like  accept  but lack any ryanmen improvement. When we choose between yonkanchan and a inside kanchan, we can reduce yonkanchan into a sankanchan without losing acceptance while increasing our ryanmen improvement draws.

Even-numbered sankanchans such as  also lack any ryanmen improvement. In most cases, we reduce it to a ryankanchan. The only case where we would keep it as is is when we are aiming for tanyao and treat it as  and  two kanchans. Therefore, we drop  from  in (63) for the faint possibility of 234 sanshoku, and we drop  pair from (64) to aim for tanyao (ゆうせー, 2022l).

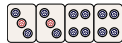
(63) 

(64) 

1.1.7 Shapes with Overlapping Acceptance

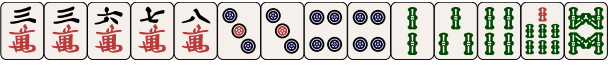
When two taatsus share the same tiles as their acceptances, our effective number of acceptance reduces. This is known as *nidouke* in Japanese, which literally means “to accept twice.” Shapes like  shares  acceptance, so only  3 kinds 12 tiles complete a mentsu out of this shape, whereas  4 kinds 16 tiles complete a mentsu out of  shape. Shapes with overlapping acceptance are not always bad or weak; there are cases where we keep them over other weak taatsus.

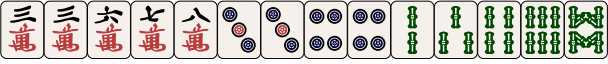
Neighboring Pairs

Shapes like  can be treated as two neighboring pairs  and  or two ryanmens  and , therefore shapes with overlapping acceptance. We can easily produce a ryanmen out of

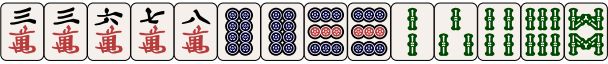
⁸Since we just cut , this could make us end up furiten.

this shape, so when pinfu is within sight, its priority is higher than weak taatsus such as kanchans, but lower than other pairs and ryanmens. Therefore, we drop one tile from  in (65), and we drop  kanchan from (66) (G・ウザク, 2019, pp. 112–113).

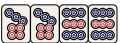
(65) 

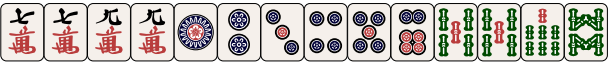
(66) 

Since (67) has no possibility of pinfu, and dropping  gives us tanyao, it becomes optimal to drop .

(67) 

Leaping Pairs

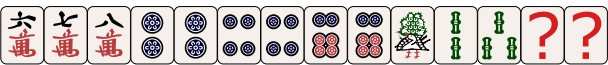
Shapes like  can be treated as two pairs  and  with a gap in between or two kanchans  and , therefore shapes with overlapping acceptance. Since drawing  gives us iipeikou, we still keep it over other kanchans and penchans, and destroy it when the rest of our hand are completed mentsus, pairs, and ryanmens (G・ウザク, 2019, pp. 114–117). In this regard, we either drop  from (68) to keep the possibility of  shanpon wait, or drop  and expect  draw which gives us complete iishanten.

(68) 

Drop  from (69). Drop  first so that we still keep the possibility of  which gives us ryanmen.

(69) 

(70) has three leaping pairs . In general case, dropping  results in  shape, which accepts  and thus have the most acceptance out of all choices, as shown in Table 29. However, the  shape gives us iipeikou if we draw , so our optimal drop depends on the rest of our hand.

(70) 

1.2 Shapes Involving Mentsus

1.2.1 Sanmenchan

Shapes like  are called *sanmenchan* in Japanese, which means “three-sided tiles.” It can be treated as either  mentsu +  ryanmen or  ryanmen +  mentsu, so drawing any of  completes this shape, hence the name.

??	Drop	Acceptance
		6/20:
		5/16:
		5/16:
		5/14:
		5/14:
		5/14:
		5/14:

Table 29: Acceptance of (70).

??	Drop
	: aim for iipeikou
	: aim for improvements of

Table 30: Tiles to drop from (70)

(71) has shape, and one may be tempted to drop to fix sanmenchan. However, dropping here from aryamen and keep it as a mentsu actually has the most acceptance and the most tiles for good-wait (including nobetan) tenpai. This is one of the cases where, in doubt, just drop aryamen.

(71) East-1, East seat, turn 4,

Drop		
Acceptance	15/51:	7/23:
Good-wait tenpai	10/26:	7/23:

Table 31: Acceptance of (71).

1.2.2 Yonrenkei

Shapes like are called *yonrenkei* in Japanese, which means “the shape of 4 consecutive numbers.” If we draw we get a pair and a mentsu; if we draw we get a ryanmen and a mentsu. It is therefore a strong shape. Depending on the rest of our hand, we may have to sacrifice this shape and fix it into a mentsu.

(72) has two yonrenkei. If we drop either of we get the most acceptance, but it is also very likely that we end up kanchan or shanpon bad-wait rii-nomi. The consists of + , which gives us iipeikou possibility. Therefore we will not touch and we opt to drop one tile from the weakest yonrenkei. Drop (G・ウザク, 2016, p. 14).

(72) East-1, West seat, turn 7, dora 

(73) has tanyao already, and with  kanchan we have tanyao dora 2, a very valuable hand. On top of that, we already have 6 blocks in our hand, the merit of producing head or ryanmen out of yonrenkei  becomes very little. Considering also the possibility of 567 sanshoku, we drop  here (G・ウザク, 2016, p. 16).

(73) East-1, East seat, turn 6, dora 

Reaching Tenpai with 3456 Shape

Generally, drop the outermost tile (ヨーテル, 2023b). If we have  and we must dropping one to reach tenpai, dropping  works in general cases. Merits of cutting the outermost tile are:

- Outer tiles generally have less deal-in rate
- Not giving other players easy time calling
- Strength of discards: a  signals that  are relatively safe, while a  gives way less information about our hand.

Exceptions are:

- Doras and sanshoku are involved
- To keep the possibility of catching kamichi's akadora
- To keep possibilities of improvements
- Inner tiles are safer according to table situation
- Early in a round

Cases involving doras and sanshoku

(74) is an open hand with . Our general policy is to drop . However, since dora is and, if we keep shape and drawing one , we can drop instead. This is called sliding and is something one must keep in mind when they have a shuntsu close to dora. Drop from (74).

(74) Dora 



If we drop $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ from (75), we will not be able to call chi on kamicha's $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ discard. If we instead keep $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, we can chi $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ with $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and discard $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. If $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is already discarded, follow the general policy and drop $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.



(76) has a slight possibility of 123 or 234 sanshoku. In this case, drop



Tipeikou-like Shapes Has Less Improvements

Dropping either of  from (77) will give us the same waits . If we drop  and keep , drawing  and dropping  gives us sanmenchan. Dropping means that we will lose this upgrade. Therefore, drop here.



Note that this is for open hands only since iipeikou shapes gives us no additional value when our hand is open. If this is a closed hand, definitely go for iipeikou since we cannot change our tiles after we riichi.

Half Suji

When another player already declared riichi and we must drop either  or  in order to take tenpai, and  is genbutsu, half suji  is safer than musuji  (みーにん, 2017, pp. 140–141). Likewise, when  is genbutsu, half suji  is safer than musuji .

The main reason why half suji is safer than musuji is because, in terms of possibility of the incomplete shape we deal in, musuji  deals into  penchan, which  does not deal into. Kanchan waits and ryanmen waits are shared by both , so  has more possible shapes to deal into than , hence higher deal-in rate.

Early in a Round

To make sure that, when we draw  back later in the game we can drop , we can drop  when it is very early in a round.

Cases of 1234 and 2345

To be able to slide when we draw middle tiles, in most cases, we drop outer tiles from  or . Other general rules such as keeping sanshoku chance or dora slide still apply.

1.2.3 23445 Shape

The **23445** shape produces a head very easily. Drawing  with  shape gives us a mentsu and a pair. When we reach tenpai with  and without a head in the rest of our hand, we can either drop , make  as head, and take  wait, or drop  and reach tenpai with  nobetan.

(78)

Drop	
Acceptance	$10/33$: $5/16$: $4/15$:

Table 32: Acceptance of (78).

When we complete the kanchan it seems that we must fix head in (78), which renders useless. However, if we drop and keep mentsu, we have no surplus tile in our hand and thus end up the most acceptance, as shown in Table 32. While it is true that drawing leaves us no choice other than tanki wait, but this is a rather small risk compared to the width of our acceptance.

1.2.4 Ryanmenkanchan

The shape is called *ryanmenkanchan* in Japanese since it has a ryanmen and a ryankanchan , thus accepting . When we draw , we have a pair, a mentsu, and a ryanmen. Since this is easily overlooked, this is a strong shape that is often ignored (G・ウザク, 2019, p. 118).

We have 6 blocks in (79) and we must drop one block. If we drop either of at that moment we have a surplus tile in our hand, and we end up 4 kinds 16 tiles of acceptance, as shown in 33. The key here is to identify the ryanmenkanchan, which makes use of all souze tiles and gives us one more kind of tiles of acceptance. Drop .

(79)

Drop	
Acceptance	$5/19$: $4/16$:

Table 33: Acceptance of (79).

(80) has 6 blocks consisting of mentsus, ryanmens, and pairs. We intend to drop either of the pair. The pinzu shape may look completed, but if we drop , there is actually ryanmenkanchan in our hand and gives us more acceptance than dropping .

(80)

--

(81) is a similar case where dropping gives us the ryanmenkanchan which makes full use of every tiles (ヨートル, 2023d).

East-1, turn 4

(81)

--

Choosing between Ryanmenkanchan and Iipeikou

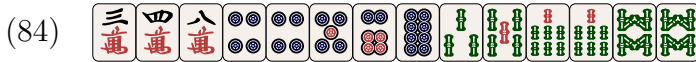
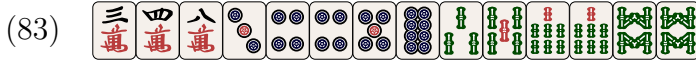
The manzu part of (82) has predecessor of iipeikou shape and ryanmenkanchan shape . Dropping either of gives us the same number of acceptance as shown in Table 34, therefore we consider the expected value of each drop. If we drop and take ryanmenkanchan, we are guaranteed pinfu. However, since pinfu tsumo only has 20fu and pinfu ron has 30fu, for the same 1 han, pinfu has slightly lower overall score than iipeikou. It is true that drop will not guarantee us iipeikou; even worse, there is a probability that we end up rii-nomi. Therefore, we leave the route to the situation where we are last and in dire need of points. Drop (in flat situation.)

(82)			
	Drop		
	Acceptance	6/19:	5/19:
	Pinfu drawn	(78.9%)	(100%)
	Iipeikou draws	(15.8%)	None

Table 34: Acceptance of (82).

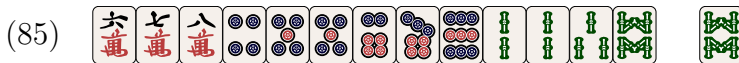
Predecessor of Ryanmenkanchan

Shapes such as and are predecessors of ryanmenkanchan; the former lacks and the latter lacks . When comparing with other 3–7 floater, we prioritize these floaters near or shape. This also applied to (83) and (84) where we are choosing between and . In both hands, we are 1 tile away from forming ryanmenkanchan, so we keep in both cases. Drop the most isolated .

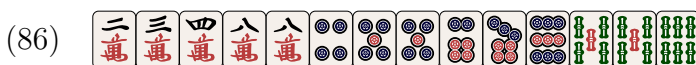


Choosing between pinfu and tanyao

For the same 1 han, tanyao has a probability of scoring higher than pinfu, and we are ready to make calls and speed up, so we prioritize tanyao over pinfu (G・ウザク, 2019, p. 119). Dropping either of from (85) leaves us no surplus tile and gives us the same 19 tiles of acceptance. Dropping gives us tanyao chance, and dropping confirms pinfu. Considering the merits mentioned earlier, we opt to drop and go for tanyao.



Likewise for (86), we drop and choose tanyao (G・ウザク, 2016, p. 10).



Dropping Ryanmenkanchan for a Better Chance of a Head

As mentioned previously in Section 1.2.3, the shape produces a head easily. To produce a head out of ryanmenkanchan, the only draw is . In hands such as (87) where we seek to confirm our head, we can drop without losing the opportunity to incorporate .

(87)

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

(90) South-2, South seat, turn 4, dora 


1.3 6 Block Theory

We have seen the term “block” here and there previously without defining its exact meaning. In this section, we will cover the term “block” thoroughly and explore the possibility of keeping multiple blocks in our hand, even cases where we have more blocks than needed.

A block is a group of tiles that we expect to become mentsu when we complete our hand. To complete a hand in mahjong using 14 tiles, we need 4 mentsus and 1 head¹⁰. In that regard, we may think that we only need to keep 5 blocks in our hand. However, when we are still far away from completion and each block does not take up too much space, we can fit 6 blocks in our hand. The method of developing our hand while keeping all 6 blocks is called “6 block theory” (G・ウザク, 2019, p. 16).

Merits of keeping 6 blocks include (G・ウザク, 2019, p. 17; ゆうせー, 2022f; ゆうせー, 2022k):

- Keeping maximal choices between yakus and waits
- Able to adapt to new table situation such as pon or kan from other players
- Dropping a tile and fixing ryanmen makes its outside wait come out more easily

Demerits of keeping 6 blocks include (G・ウザク, 2019, p. 18; ゆうせー, 2022f):

- Keeping 6 blocks leaves us no room for follow-up tiles for each block, thus harder to reach complete iishanten
- The moment we drop 1 block to take 1-away, the destroyed block leaves us 1 surplus tile, thus making our acceptance narrower and lowers our defense

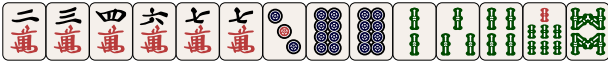
1.3.1 Standard of Keeping 6 Blocks

Our policy regarding block number in our hand is (G・ウザク, 2019, p. 22–23; ヨーテル, 2023c; ゆうせー, 2022f; ゆうせー, 2022k):

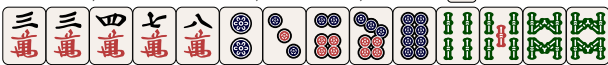
- Keep 6 blocks when
 - Going straightforward leaves us high probability of bad wait
 - Blocks involve hand value and yaku
 - There is no huge difference between blocks
- Keep 5 blocks when
 - There is huge difference between blocks
 - We prioritize the speed of reaching tenpai
 - Our hand is open and we want pairs for pon calls
 - We prioritize defense

¹⁰Except for kokushi musou and chiitoitsu.

Keep 6 blocks if going straightforward with blocks we currently have leaves us bad waits or cheap hand (G・ウザク, 2019, p. 23). For (91), if there is no difference in player points, drop  and take complete iishanten. However, if we are far behind other players, keeping  gives us a slight chance of sanshoku and dropping  confirms pinfu. This is the case where even a single tile can be treated as a whole block.

(91) 

When there is no discernible difference between taatsus, we can delay our decision by keeping 6 blocks (ヨーテル, 2023c). (92) has 6 taatsus, and dropping any of  leaves us 5 blocks. We can also fix  into a ryanmen and progress with 6 blocks.

(92) East-1, South seat, turn 3, dora 


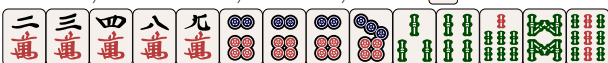
We can drop  from (93) and keep 6 blocks, or drop  and reduce it to 5 blocks¹¹. If we drop , drawing  will not destroy sanshoku, and we keep the possibility of drawing another copy of . If any other ryanmen gets filled up before we complete , we can drop  and aim for pinfu. This is the case where keeping 6 blocks gives us more choices between yakus (ゆうせー, 2022f).

(93) East-1, South seat, turn 3, dora 


(94) has a penchan that is clearly weaker than kanchan, so we drop  (石橋伸洋, 2015, p. 21).

(94) 

(95) has  shape which can be treated as 2 blocks  + . If we reduce it to  1 block, we have the widest acceptance of 7 kinds 24 tiles. However, if we don't draw any of  9 tiles (1 copy of  is the dora indicator) first, we end up a weak penchan wait. We therefore drop . If we later draw another copy of dora , we can drop  (G・ウザク, 2016, p. 16).

(95) East-1, East seat, turn 7, dora 


Being able to pon is crucial for open hands (ヨーテル, 2023c). We just called pon  in (96). Since we can change tiles and waits with open hands, having tiles following taatsus becomes more important and contributes to our flexibility. If (96) develops into (97), calling chi  gives us two ryanmens, drawing  also improves our hand into ryanmens. Therefore, drop either of  depending on table situation.

¹¹If we drop  we lose pinfu and sanshoku becomes undetermined. Do not drop .

(96)

- By keeping two pairs, we keep the possibility of their ryanmen improvements and therefore the possibility of complete iishanten
- If our pairs are outside, they come out more likely, especially for honor tile pairs
- Two pairs means that our waits becomes *atosuji*¹² twice likely
- Pairs do not use neighboring tiles, thus making them no-chance¹³ more likely

In terms of ryanmen improvement and complex taatsu improvement, outer kanchan and one pair are weaker than two pairs of number 2 (or 8). When we have two pairs, we keep ryanmen improvement and complex taatsu improvement of each pair. On the contrary, when we only have a pair and a kanchan, we must keep the pair as our head, so we lose its improvement; our only improvement becomes the improvement of kanchan. Table 36 lists acceptance and all improvement of  (a kanchan and a pair) and  (pairs of number 2 and 8.)

Shape		
Acceptance	1/4: 	2/4: 
Ryanmen improvement	1/4: 	2/8: 
Ryankanchan improvement	1/4: 	
Kanchan-toitsu improvement	None	2/8: 

Table 36: Comparing  and .

Even for pairs like  which lack any ryanmen improvement, due to them being the outmost number tiles and twice likely becoming atosuji than a kanchan, they are decent as final waits. In this regard, the only cases where we would keep kanchan is when we have yakus such as tanyao or sanshoku involved, or when table situation makes pairs less favorable than kanchan.

When we decide to reduce our hand to 5 blocks, keep safe tiles over the block to drop (お知らせ, 2020, pp. 152–153). At the moment of drawing  in (102), we have 6 blocks. We will not touch yakuhai pairs that give us value and speed; in that regard, we have no incentive to keep . Instead of keeping  and maintaining 6 blocks until we complete another block, we must drop it immediately to make room for  safe tile. Drop .

(102) South-2, East seat, turn 4, dora 


Comparing Similar Shapes

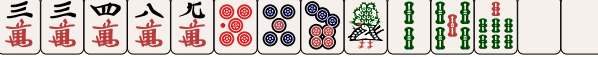
Even for shapes in the same class, since their acceptances are same (without table condition,) their differences lie in their improvement and shapes around them. In particular, we must also take into consideration whether or not we can still incorporate the tiles their accept when we drop the taatsu and its accepted tiles backfires. For each class of shapes, we have following merits (お知らせ, 2020, pp. 152–153; ゆうせー, 2022j):

¹²When our discards after riichi makes our wait suji, it is called atosuji. The part *ato* means “behind, after” in Japanese.

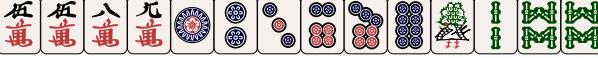
¹³If there are 4 copies of , it makes  no-chance since the ryanmens waiting on  or  requires 1 copy of . It makes  kanchan also impossible. It, however, does not negate the possibility of  or  shanpon wait.

1. (Kanchan, penchan) remaining number of acceptance
2. (Ryanmen, penchan) follow-up of acceptance
3. (Kanchan) follow-up of ryanmen-improvement, ryankanchan-improvement draws
4. (Toitsu) whether or not we can slide towards dora

There are two penchans  and  in (103). To prepare for the moment we pon , we must keep another pair , so we cannot drop . In this hand, after dropping , its acceptance  can still be incorporated by its neighboring kanchan, which forms  ryankanchan. On the other hand, drawing  after dropping  is useless; it is just a floater with furiten risk. In other words, dropping  loses the least. Drop  (ゆうせー, 2022j).

(103) East-1, North seat, turn 3, 


(104) is also choosing between  and . Since we have  pair, the acceptance  of penchan  gives us  kanchan-toitsu, and its ryanmen-improvement draw  gives  ryanmen-toitsu. When there is no other information available to us, judging on how we can still incorporate accepted tiles and improvement draws,  becomes the weakest penchan. However, when  is once cut, or when we immediately draw  after dropping , we still keep manzu and drop  (ゆうせー, 2022j).

(104) East-1, East seat, turn 2, dora 


(105) is choosing between  and . With  shape, when we drop  and draw , we can still slide towards  and drop . Drop  (ネマタ, 2014d).

(105) 

(106) is choosing between  and . With  shape, when we draw another dora , we can drop  and maintaining a head in our hand. Therefore, drop  (G・ウザク, 2016, p. 20).

(106) East-1, West seat, turn 7, dora 


2 Bibliography

Chiba, Daina (2020) *Riichi Book I*.

chirno (2008) 「牌効率について (リャンカントイツやサンカンチャン等)」, URL : <https://chirno.hatenadiary.org/entries/2008/05/18> (アクセス日: 2023 年 7 月 29 日).

- David (2018a) “[Houou Lab] Looking at Tenhou ASAPIN’s Calling Part 2,” URL: <https://osamuko.com/hououlab-asapin2/>, accessed on 2024.
- (2018b) “[Houou Lab] Looking at Tenhou ASAPIN’s Calling Part I,” URL: <https://osamuko.com/hououlab-asapin1/>, accessed on 2024.
- Equim-chan (2023) “mjai-reviewer/faq.md at master · Equim-chan/mjai-reviewer,” URL: <https://github.com/Equim-chan/mjai-reviewer/blob/master/faq.md>, accessed on 2023.
- Feng (2018) *Intermediate Riichi Strategy*.
- G・ウザク (2016) 『麻雀 傑作「何切る」300 選』, 三才ブックス.
- (2019) 『ウザク式 麻雀学習 牌効率』, 三才ブックス.
- manboo (2013) 「イーシャンテンの牌理 (1) | 無料ネット麻雀戦術書」, URL: <http://beginners.biz/pairi/pairi13.html> (アクセス日: 2023 年 5 月 6 日).
- nisi (2014a) 「とりあえず麻雀研究始めてみました ノーチャンスワンチャンス」, URL: <http://epsilon69399.blog20.fc2.com/blog-entry-117.html> (アクセス日: 2023).
- (2014b) 「真ん中無筋カンチャンリーチのみの是非」, URL: <http://epsilon69399.blog20.fc2.com/blog-entry-164.html> (アクセス日: 2023 年 5 月 6 日).
- noprops (2022) 「麻雀 一択何切る」, URL: <https://play.google.com/store/apps/details?id=info.mygames888.nanikiru&hl=ja> (アクセス日: 2024).
- tomo sann (2022) 「麻雀の攻守ともに重要!!～一向聴 (イーシャンテン) の種類を考える～ 麻雀大好きともさんブログ」, URL: https://majann-daisuki.com/majan-ishanten/?utm_source=rss&utm_medium=rss&utm_campaign=majan-ishanten (アクセス日: 2023 年 5 月 6 日).
- お知らせ (2020) 『世界最強麻雀 AI Suphx の衝撃』, マイナビ出版.
- とつげき東北 (2021) 「新 科学する麻雀」, URL: <https://hobbyjapan.co.jp/mahjong/data.html> (アクセス日: 2024).
- ひい (2005) 「2 - 0 4 牌効率」, URL: <http://www.ix3.jp/hiii/02mahken/2-04paikouritsu.htm> (アクセス日: 2023 年 5 月 28 日).
- みーにん (2017) 『「統計学」のマーじゃん戦術』, 竹書房.
- みーにん・梶本琢程 (2019) 『知るだけで強くなる麻雀の 2 択』, 竹書房.
- ゆうせー (2020a) 「どのターツが不要かを【的確に】選ぶ基準～“配牌からの”牌効率 第 2 回～リーチを目指せ! 麻雀初心者向け講座!」, URL: <https://www.youtube.com/watch?v=EWIZuxUXovo> (アクセス日: 2024).
- (2020b) 「孤立浮き牌とペンチャンターツ、どちらを切るか!? 比較&解説!【ライバルに差をつけよう!】～明日から使える麻雀戦術講義!～」, URL: <https://www.youtube.com/watch?v=Pjf0RhPbo4Y> (アクセス日: 2024).

- (2020c) 「数牌トイツと愚形ターツの、明日から使える【法則】紹介！ “配牌からの” 牌効率 第 3 回～リーチを目指せ！麻雀初心者向け講座！」, URL : <https://www.youtube.com/watch?v=VCrMiHnieTc> (アクセス日: 2024).
- (2021a) 「24667 などの連続形の強さと【意外と知られていない】捌き方の極意」, URL : <https://www.youtube.com/watch?v=rm4kzJuW6B8> (アクセス日: 2024).
- (2021b) 「【#M リーグ】令和麻雀の必須事項！完全イーシャンテン先切りマニュアル～明日から使える麻雀戦術講義！22 回目～」, URL : <https://www.youtube.com/watch?v=fvKnLKqewpM> (アクセス日: 2023 年 11 月 4 日).
- (2021c) 「データの衝撃があなたを襲う！今すぐに取り入れたい 1335 の捌き方徹底講座！明日から使える麻雀戦術講義！24 回目～」, URL : <https://www.youtube.com/watch?v=VAyF-DODGcg> (アクセス日: 2024).
- (2021d) 「初・中級者が間違えやすい何切るを解いて【重要戦術】を 3 つ身につけよう！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 32 回～」, URL : <https://www.youtube.com/watch?v=8BdKtyk195o> (アクセス日: 2024).
- (2021e) 「初心者の大半が知らない 11355 をさばくコツを 10 分強でマスター！～初心者から上級者まで！～リーチを目指せ！ “配牌からの” 牌効率講座！第 14 回」, URL : <https://www.youtube.com/watch?v=cR4bJxaShX0> (アクセス日: 2023 年 7 月 29 日).
- (2022a) 「99%の初心者が間違える 2357 何切る全 4 問！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 41 回～」, URL : <https://www.youtube.com/watch?v=631L6QcUEAw> (アクセス日: 2024).
- (2022b) 「あまりにも簡単すぎてガチで驚きます。3568 マスター講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 46 回～」, URL : <https://www.youtube.com/watch?v=Gfksmav7wEs> (アクセス日: 2024).
- (2022c) 「こういうのを分からないままにするのが一番ダメです。34468 に 3 を引いたときは？～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 45 回～」, URL : <https://www.youtube.com/watch?v=Ad5mgfoGfJ0> (アクセス日: 2024).
- (2022d) 「まだ間違った牌を切ってるんですか...？112246 のさばき方徹底講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 47 回～」, URL : <https://www.youtube.com/watch?v=iEtFXGbAHE8> (アクセス日: 2023 年 5 月 6 日).
- (2022e) 「永久保存版 誰でも赤 578 がさばけるコツを説明！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 43 回～」, URL : <https://www.youtube.com/watch?v=5Lx03nDGk0w> (アクセス日: 2023 年 5 月 6 日).
- (2022f) 「覚えたら麻雀が変わる【ブロック使い分け】完全マスター講座！～明日から使える麻雀戦術講義！32 回目」, URL : <https://www.youtube.com/watch?v=oq2ntvWUI1M> (アクセス日: 2024).
- (2022g) 「苦節 20 年。やっと発見した 6889 の法則を紹介します。～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 48 回～」, URL : <https://www.youtube.com/watch?v=4jeri6yDIkA> (アクセス日: 2023 年 5 月 6 日).

- (2022h) 「今日から悩まずに切れる！35689 のさばき方徹底講座！～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 35 回～」, URL : https://www.youtube.com/watch?v=_V000khUCw0 (アクセス日: 2024).
- (2022i) 「衝撃の結果が?! 一生使える基準付き！134 ドラ 1 は何を切る?～役牌ボン編～明日から使える麻雀戦術講義！36 回目」, URL : <https://www.youtube.com/watch?v=VxUirDseXPw> (アクセス日: 2023 年 5 月 6 日).
- (2022j) 「多くの上級者が使いこなす「裏目フォロー打法」をマスターしよう!～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 36 回～」, URL : <https://www.youtube.com/watch?v=cB7k0t0N41Y> (アクセス日: 2024).
- (2022k) 「誰も教えてくれない 6 ブロック打法の弊害とは?～明日から使える麻雀戦術講義！35 回目」, URL : <https://www.youtube.com/watch?v=Dn29q4DfHaE> (アクセス日: 2024).
- (2022l) 「土田浩翔は正しかった！極悪形 2468 の攻略法!～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 42 回～」, URL : <https://www.youtube.com/watch?v=2chbGHCzpnY> (アクセス日: 2023 年 5 月 6 日).
- (2023a) 「こういうのを分からないままにするのが一番ダメです。孤立 28 と役牌を選ぶ基準! (手なり編) 初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 54 回～」, URL : <https://www.youtube.com/watch?v=yi6-g0ws6Zk> (アクセス日: 2024).
- (2023b) 「知らないと絶対に損している 126 から 1 枚選ぶコツ～初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 53 回～」, URL : <https://www.youtube.com/watch?v=Go3CHktsAV0> (アクセス日: 2024).
- (2024a) 「99%の初心者が間違えている 19 牌と役牌のガチで効果的な切り順マスター講座！初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 56 回」, URL : <https://www.youtube.com/watch?v=c2bGmxHWKH0> (アクセス日: 2024).
- (2024b) 「なぜ中級者のミスが多発しているのか、動画で説明します。13r5 のコツとは?! 初心者から上級者まで！リーチを目指せ！ “配牌からの” 牌効率講座！第 57 回」, URL : <https://www.youtube.com/watch?v=0zLZJMZO1ro> (アクセス日: 2024).
- クリアレイン (2023) 「【麻雀】知らないで絶対にあがり率が下がる「2457」の扱い方【牌効率】」, URL : <https://www.youtube.com/watch?v=N6eK0Ht8WNo> (アクセス日: 2024).
- (2024) 「【麻雀】知らないで絶対に損する「34468」の扱い方【牌効率】」, URL : <https://www.youtube.com/watch?v=CqVrnEQStdg> (アクセス日: 2024).
- ツケマイせんせい (2014) 「シャボかカンチャン論争 | ツケマイせんせいの哲学する麻雀」, URL : <https://ameblo.jp/nyanyami2011/entry-11764202490.html> (アクセス日: 2023 年 5 月 25 日).
- ネマタ (2014a) 「現代麻雀技術論 - 牌効率論 10・外カンチャンターツ」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section010.html> (アクセス日: 2023 年 5 月 6 日).

- (2014b) 「現代麻雀技術論 - 牌効率論 11・内カンチャンターツ」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section011.html> (アクセス日: 2023 年 5 月 6 日).
- (2014c) 「現代麻雀技術論 - 牌効率論 13・トイツ落とし」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section013.html> (アクセス日: 2023 年 7 月 28 日).
- (2014d) 「現代麻雀技術論 - 牌効率論 45・2 面子余剰牌形」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section045.html> (アクセス日: 2023 年 5 月 20 日).
- (2014e) 「現代麻雀技術論 - 牌効率論 47・2 面子余剰牌形か完全形の選択 1」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section047.html> (アクセス日: 2023 年 5 月 20 日).
- (2014f) 「現代麻雀技術論 - 牌効率論 56・2 シャンテン戻し」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section056.html> (アクセス日: 2023 年 10 月 28 日).
- (2014g) 「現代麻雀技術論 - 牌効率論 5・複合ターツ 1」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section005.html> (アクセス日: 2023 年 5 月 6 日).
- (2014h) 「現代麻雀技術論 - 牌効率論 6・複合ターツ 2」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section006.html> (アクセス日: 2023 年 5 月 6 日).
- (2014i) 「現代麻雀技術論 - 牌効率論 9・ペンチャンターツ」, URL : <http://yabejp.web.fc2.com/mahjong/tactics/chapter01/section009.html> (アクセス日: 2023 年 5 月 6 日).
- (2016) 「もっと勝つための現代麻雀技術論 第 133 回「別の面子、雀頭、面子候補を崩す鳴き」」, URL : <https://mj-news.net/column/method/2016021434639> (アクセス日: 2024).
- ヨーテル (2022) 「【麻雀講座】もう迷わない! 孤立ドラ役牌の扱い方【天鳳位】」, URL : <https://www.youtube.com/watch?v=HQphpEWog4s> (アクセス日: 2024).
- (2023a) 「【麻雀講座】2 枚切れの愚形でリーチするか?【天鳳位】」, URL : <https://www.youtube.com/watch?v=Z7QmW6Rn9Eo> (アクセス日: 2024).
- (2023b) 「【麻雀講座】3 4 5 6 の 3 と 6、適当に切ると痛い目見ます【牌効率/天鳳位】」, URL : https://www.youtube.com/watch?v=6AA8f_vZ2pM (アクセス日: 2024).
- (2023c) 「【麻雀講座】5 ブロックと 6 ブロック、2 つの道の選択基準【天鳳位】」, URL : https://www.youtube.com/watch?v=F0_KVS1oMV4 (アクセス日: 2024).
- (2023d) 「【麻雀講座】下手に見られないために、上位互換を選ぼう【天鳳位】」, URL : <https://www.youtube.com/watch?v=mPJNh8NcU> (アクセス日: 2024).
- (2023e) 「【麻雀講座】配牌で役牌対子が 2 つあるときの進め方【天鳳位】」, URL : <https://www.youtube.com/watch?v=nVnd5xBs-po> (アクセス日: 2024).

- (2024) 「【麻雀講座】 1 3 4 の 1 を雑に切ってはいけません【牌効率/天鳳位】」, URL : <https://www.youtube.com/watch?v=ai-nHr4Eg14> (アクセス日: 2024).
- 綾鷹 (2020) 「愚形リーチのみ」, URL : https://note.com/ayataka_tell/n/n277c956cf9de (アクセス日: 2023 年 5 月 8 日).
- 筋トレプレーヤー (2024) 「麻雀の対リーチ放銃点の期待値に関する考察」, URL : <https://www.withdrawal-civilservice.com/entry/2024/03/01/203018> (アクセス日: 2024).
- 国税庁長官官房企画課 (2023) 「民間給与実態統計調査」.
- 渋谷難波 (2023) 「【渋谷ハル】守備力を上げる！手出し・ツモ切りを覚えるより大事！ペンチャン・カンチャンなど愚形リーチへの放銃を減らす方法！【雀魂/M リーグ/KADOKAWA サクラナイト/渋谷難波切り抜き】」, URL : <https://www.youtube.com/watch?v=H6fR3tYGg0w> (アクセス日: 2024).
- (2024) 「ノータイムで を切れたらウザク式卒業!? 最終形の強さを求める問題！【神域リーグ 2024/渋谷難波切り抜き】」, URL : <https://www.youtube.com/watch?v=Pqnd1iJK3Mc> (アクセス日: 2024).
- 石橋伸洋 (2015) 『黒いデジタル麻雀～現代流データ戦術を斬る～』, マイナビ出版.
- 多井隆晴 (2021) 『麻雀無敗の手筋』, 竹書房.
- 滝沢和典 (2015) 『滝沢和典の麻雀読みの公式』, マイナビ出版.
- 土井泰昭 (2021) 『シンプルにして最重要! 麻雀 押し引きを決める手牌価値』, マイナビ麻雀 BOOKS.
- 独歩・かにマジン・しゅかつ (2017a) 『麻雀・序盤の鉄戦略: 3 人の天鳳位が出す究極の結論』, マイナビ出版.
- (2017b) 『麻雀・鉄押しの条件: 3 人の天鳳位が出す究極の結論』, マイナビ出版.
- 平澤元気 (2019) 『鳴きの教科書』, マイナビ出版.
- (2021) 『通せる牌が倍増! 麻雀鳴き読みドリル』, マイナビ出版.
- (2023) 「上級者は全員やってるけど当然すぎてあまり説明してくれない食い伸ばしのテクニック」, URL : <https://www.youtube.com/watch?v=MAV2AgoMU-w> (アクセス日: 2024).
- 鳳東のうさぎさん (2020) 「邪魔ポンを使いこなす」, URL : <https://note.com/houtooooooooon/n/n06b7869e19f3> (アクセス日: 2024).
- 堀内正人 (2015) 『神速の麻雀 堀内システム 51』, 洋泉社.